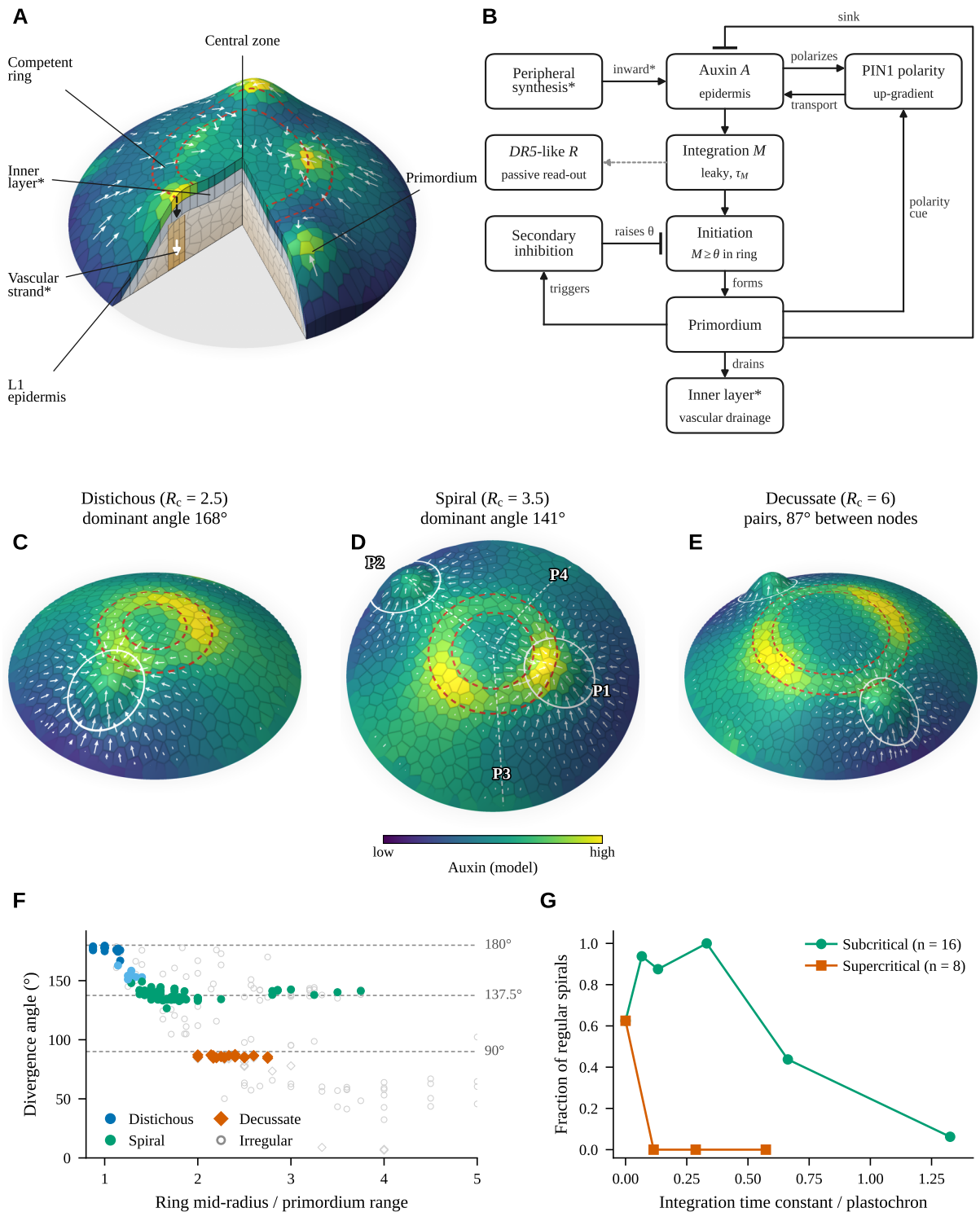

A
Central zone
Competent ring
Inner layer*
Primordium
Vascular strand*
L1 epidermis
B
sink
Peripheral synthesis*
inward*
Auxin A
epidermis
polarizes
transport
PIN1 polarity
up-gradient
DR5-like R
passive read-out
Integration M
leaky, τM
polarity cue
Secondary inhibition
raises θ
Initiation
M ≥ θ in ring
triggers
forms
Primordium
drains
Inner layer*
vascular drainage
Distichous (Rc = 2.5)
dominant angle 168°
Spiral (Rc = 3.5)
dominant angle 141°
Decussate (Rc = 6)
pairs, 87° between nodes
C
D
E
P2
P4
P1
P3
low
high
Auxin (model)
F
G
Divergence angle (°)
180°
137.5°
90°
Distichous
Spiral
Decussate
Irregular
Ring mid-radius / primordium range
Fraction of regular spirals
Subcritical (n = 16)
Supercritical (n = 8)
Integration time constant / plastochron